

Practical No.10

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199.

Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

❖ CODE

 /c/Users/acom/Desktop/OS_CD24037

```
GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>

int main() {
    int request[8] = {98,183,41,122,14,124,65,67};
    int visited[8] = {0};
    int head = 53;
    int total = 0;
    int n = 8;

    for(int i=0;i<n;i++) {
        int min = 9999;
        int index = -1;

        for(int j=0;j<n;j++) {
            if(visited[j]==0) {
                int distance = abs(head - request[j]);
                if(distance < min) {
                    min = distance;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        printf("Head moves from %d to %d = %d\n", head, request[index], min);

        total += min;
        head = request[index];
    }

    printf("\nTotal Head Movement = %d\n", total);

    return 0;
}
```

❖ Output

 /c/Users/acom/Desktop/OS_CD24037

```
acom@DESKTOP-50T9G9Q MSYS ~  
$ cd /c/Users/acom/Desktop/OS_CD24037  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ nano sstf1.c  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ gcc sstf1.c -o sstf1  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ ./sstf1  
Head moves from 53 to 41 = 12  
Head moves from 41 to 65 = 24  
Head moves from 65 to 67 = 2  
Head moves from 67 to 98 = 31  
Head moves from 98 to 122 = 24  
Head moves from 122 to 124 = 2  
Head moves from 124 to 183 = 59  
Head moves from 183 to 14 = 169  
  
Total Head Movement = 323  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ |
```